



An inductive typology of egocentric networks with data from the Socio-Economic Panel

Bastian Laier^{a,*}, Marina Hennig^a, Stefan Hundsdorfer^b

^a Jakob-Welder-Weg 12, 55128 Mainz, Germany

^b Spitalgasse 2, 1090 Wien, Austria

ARTICLE INFO

Keywords:

Egocentric networks
Clustering
Machine learning
Random Forests

ABSTRACT

In this article we apply Random Forests to data from the German Socio-Economic-Panel (SOEP), creating an inductive typology of egocentric networks. Using an earlier application of the machine learning algorithm as a guideline, we are putting the wider applicability of the method to the test by using data not exclusively constructed for network analysis and focusing on core networks of respondents. Applying Random Forests promises more differentiated compositional typologies than currently used. Overcoming the prevailing approach to typologies by creating a larger number of finer-grained types can improve network-oriented life course analysis. We started our analysis by selecting 44 descriptors of 8341 respondent networks and reduced this number by combining most of them to indices representing basic dimensions of egocentric networks. The descriptors are used in hierarchical clustering to derive clusters that are then predicted with Random Forests to evaluate the reliability of this approach. We can identify stable core differentiators underlying typology construction but are also confronted with variation in the resulting types. While some of this can be traced back to the distinct socio-demographics of the respective populations, some is the result of measurement disparities. Against the backdrop of the presented results, we discuss the relationship between a deductive approach of network sociology, dedicated to theory-testing and inductive typologies of egocentric networks within a framework of life course analysis.

1. Introduction

In the social sciences, we are often interested in typologies to reduce the complexity of our data. A denumerable number of characteristic core network types is not only advantageous to structure the analysis of large datasets but can also serve as an interim step in the process of theory formation. People's core networks show us who they rely on in times of need, who they socialize and confer with. They tell us something about their relation towards proximate others and their embeddedness into wider society. When communicating social scientific results to the public, politicians or social service providers, typologies are indispensable. They often represent archetypes that are easily understood by everyone and can be related to. But there is another crucial application of typologies that is seldom mentioned: As a heuristic in the generation and refinement of research frameworks, a part of the research process that is rarely made explicit. Researchers in the field of social network analysis have identified various typologies of egocentric networks. They have been created for distinct reasons and were focusing on different aspects (Bidart et al., 2018; Fiori et al., 2006, 2008; Giannella and Fischer, 2016; Litwin, 1995a; Maya-Jariego, 2021;

Park et al., 2015, 2018; Vacca, 2020; Wenger, 1989). Generally, structural, compositional, and mixed typologies can be discerned (McCarty et al., 2019; Vacca, 2020).

Due to progress in network data acquisition and theoretical advancements network typologies focusing on purely structural aspects can now be derived (Bidart et al., 2018; Maya-Jariego, 2021; Vacca, 2020). Variations in network composition are particularly relevant for research questions in life course research, family sociology and the micro-foundational analysis of demographic change. These research areas, which are of particular interest for us in this article, often rely on nationwide panels and cross-national surveys, containing only limited explicit network data. We see a demand for enhanced techniques of typology creation in these research areas. Therefore, we focus entirely on compositional network typologies. So far, most attempts at compositional typology construction can be attributed to gerontology and social support research (Fiori et al., 2006, 2008; Guadalupe and Vicente, 2020; Hlebec and Kogovšek, 2013; Litwin, 1995a; Park et al., 2015; Szabo et al., 2018; Wenger, 1991). Variables used to derive such typologies often consist of proportions of friend and family confidants and

* Corresponding author.

E-mail address: laier@uni-mainz.de (B. Laier).

<https://doi.org/10.1016/j.socnet.2022.07.005>

neighbors (Kogovšek et al., 2013), measures for size and contact frequency for different categories of alters, as well as perceived closeness (Park et al., 2018). Sometimes measures of community involvement (Wenger, 1991), geographical proximity, the duration of ties (Litwin, 1995a) and the sources of emotional and instrumental support (Fiori et al., 2008; Wenger and Shahmasebi, 1991) are added.

Due to the use of a limited set of variables and traditional forms of cluster analysis, repeatedly only a small number of types are produced. Often, only four to six different types are reported (Fiori et al., 2008; Litwin, 1995a; Park et al., 2015; Wenger and Shahmasebi, 1991). These typologies usually consist of a family-centered, a friend-centered, a diverse, and a restricted type (Fiori et al., 2008; Litwin, 2001; Park et al., 2018). At times the dimension of community involvement or neighborhood integration is included (Guadalupe and Vicente, 2020; Szabo et al., 2018; Wenger, 1991). What seems parsimonious at first glance, appears – under closer scrutiny – as a typology too superficial to cover the complexity of social reality of the broader population. Increasing availability of detailed information on the composition of networks in the last decade did not always lead to a subsequent improvement of the statistical tools to analyze them. Another issue hindering the progress of compositional or mixed network typologies, is their insensitivity towards the location in the life cycle. Due to the widespread application of network analysis in epidemiology and gerontology, typologies are predominantly construed for networks of older people (Fiori et al., 2006, 2008; Guadalupe and Vicente, 2020; Litwin, 1995a; Park et al., 2015; Wenger, 1989). Only in rare cases are they created for the entire adult population of a region or country (Giannella and Fischer, 2016; Park et al., 2018). Owing to the changing relevance of social support or friendship ties over the life course, resulting typologies cannot be generalized to the entire population. It is not plausible, that if four to six types are needed to cover the variations of networks of the elderly, the same amount should suffice for the entire age range. Giannella and Fischer's (2016) approach is the only one known to us, which is both representative for the whole age range of a region (Northern California) and is at the same time using network data detailed enough to be able to produce a refined compositional typology. Applying Random Forests (RF) results in a larger number of types, in the case of Giannella and Fischer's study (2016) 11 are found. Certainly, more types do not necessarily create better insights. However, we maintain, that the restriction to the dimension of family-, friend- and community orientation, which pervades network typologies in the social support or gerontology literature, leads to an oversimplification of network types and conceals important differences in network characteristics.

To uncover patterns which have been going unnoticed before, RF generated typologies may serve as a relevant exploratory tool. When applying the procedure, more fine-grained network types than previously found can be expected. Its inductive nature allows for computations without detailed prior knowledge about the descriptive properties of the variables used, which is particularly helpful when dealing with a multitude of them. The question we now ask ourselves is, whether similar differentiators can be identified inductively when RF is applied to nationwide representative data, that is of more recent origin than Giannella and Fischer's and is covering only core networks of respondents.¹ Can these differentiators structure the data in an interpretable manner, so consistent and differentiated network types applicable to the frameworks of life course and family sociology can be derived?

¹ While this restriction on core networks does not allow for the analysis of complex forms of embeddedness, 'dormant' (Vacchiano and Spini, 2021) or weak ties (Granovetter, 1973), the central notion of social support – in the forms of expressive and instrumental resources provided by strong ties of ego, as described in Lin's network theory of social capital (Lin, 1999) – can be assessed. These strong ties can furthermore be attributed to several contexts or foci of social life (Feld, 1981).

We know from Giannella and Fischer's experience, that RF is a method that reacts sensitively to changes in data preparation and setup (2016: 22), so we are curious whether robust results can be obtained. To address these questions, we are applying RF to analyze data from the Socio-Economic Panel (SOEP) in Germany. If the results are on one hand comparable to Giannella and Fischer's even when network data is limited, and on the other hand are sufficiently different to cover cultural variations, as well as differences in the social environments of the respondents, this could be considered a steppingstone towards improved typologies in large-scale surveys.

2. Testing the viability of the RF approach for a large dataset

The main goal of the article is to test the assumption that RF as an inductive approach can identify similar core differentiators for social network types, even when applied to large datasets not exclusively designed for network analysis. To find out whether homologous patterns to those in Giannella and Fischer's typology (2016) can be retrieved, using a much bigger sample from a different cultural setting and slightly different variables while focusing on the core networks of the respondents, we apply their approach to the German SOEP. We expect a certain stability in the resulting types, due to a limited variation of social networks and support relations between western countries (e.g. Freeman and Ruan, 1997). We simultaneously expect variability due to differences in the age structure, proportion of migrants and proportion of people with migration background as well as differences in religious activities like church attendance (PEW Research Center, 2016) between the US and Germany. This leads us to a subordinate goal: we want to contribute to the discussion on the development of network typologies and their purpose in the research process. We will assess the potential of using inductive typologies as a heuristic in a larger context of sociological theory building. We are especially concerned with the viability of the method and its potential value within a framework of network-focused life course analysis (Bernardi et al., 2019; Settersten, 2015; Vacchiano and Spini, 2021). So far existing typologies have been too crude to allow the analysis of subtle changes of core networks in the life course. More differentiated types could not only help analyze such shifts in core network types, they could also be a valuable tool to assess their respective impact on factors like well-being, physical and psychological health, strengthening a decidedly network-focused approach on life course research (e.g. Vacchiano and Spini, 2021; Settersten, 2015).

In accordance with the method proposed by Giannella and Fischer (2016), we first selected relevant descriptors from the SOEP to describe egocentric networks. We end up with 44 variables (see the supplement for descriptions) that are at least to a certain degree comparable to those in the initial study and that are created using standard questions employed when eliciting egocentric networks (like discussing important personal matters (Burt, 1984) or personal worries (Fischer, 1982b)). We then used these variables in the machine learning algorithm Random Forests by Breiman (2001) to calculate a proximity matrix, which we then applied in a second step to cluster the data. As stated by Giannella and Fischer (2016) this technique is relatively new for sociologists, and its usefulness for the analysis of social science data is currently under scrutiny. Although we had no prior knowledge of this technique, we were able to conduct the analysis following the detailed descriptions of Giannella and Fischer (2016) in their supplement.

3. Data and methods

We use a large German survey, the SOEP (SOEP v33).² This is a representative panel survey of private households in Germany (approximately 12,000), conducted annually since 1984 with the same sample.

² The SOEP is conducted by the German Institute for Economic Research and offers English-language manuals and data descriptions on its website (https://www.diw.de/sixcms/detail.php?id=diw_01.c.592138.en).

Table 1

Comparison of Socio-Economic Panel (SOEP) and Northern California Community Study (NCCS).

Source: Own calculations with the [Socio-Economic Panel \(SOEP\) \(2020\)](#) and [Fischer \(1982a\)](#).

	SOEP	NCCS
Survey year	2010–2011	1977–1978
Age restriction	respondents had to be 18 years or older	respondents had to be 18 years or older
Sample	random sample of approximately 12.000 households in Germany	random sample of English-speaking adults living in predominantly white urban places in northern California
Sample size	28,733	1,050
Number of name generators	5	10
Alter-alter relations	not available	for a partial network, it was asked whether the Alteri felt strongly connected to each other
Generator restrictions	number of alteri limited to five	number of alteri unrestricted
Network size (mean)	4.2	18.5
Density (mean)	not available	0.6
IQV_role (mean)	0.66	not available
Proportion of kin (mean)	0.77	0.42

Since the inception of the household survey, several supplements and subsamples have been added. With the addition of several subsamples as well as supplemental surveys, an ever-increasing capacity for analysis is being achieved. The data cover more and more life stages, and by interviewing an increasing number of people, greater representativeness is achieved ([Wagner et al., 2007](#), 152 p.). Among the extensions of the survey is the topic of social network involvement, which is collected every five years. We use the 2011 data because this was the first time five people could be named on five name generators and because important variables were not collected in other survey waves, e.g. the extent of social or cultural activities. Respondents were between the ages of 18 and 96. Since there are few household surveys with such numbers of cases on network involvement that allow linkage to extensive information on the life histories and life course-related behavior of 28,733 respondents, we used this dataset for the analyses.³ Although our analysis used the same procedure as [Giannella and Fischer \(2016\)](#), the underlying datasets differ in important ways that should be kept in mind when interpreting our results. For this reason, [Table 1](#) provides a comparative overview of key differences between the two datasets.

In the 2011 wave of the SOEP, the egocentric networks of respondents were collected. Several name generators were used ([Wolf, 2010](#)), for example who the respondents turn to in order to get support or talk about personal matters. In addition, name interpreters were used to gather some basic information about these connections. As a first step, we have selected variables that are expected to be able to describe egocentric networks. We then constructed variables which are for the most part equivalent to the data used by [Giannella and Fischer \(2016\)](#). The constructed variables mainly represent the number of different types of persons the respondents named. For example, we constructed a variable containing the number of kin that were named as a person of trust. We have also constructed descriptors such as an index of qualitative variation (IQV_role) to indicate how diverse respondents' personal networks are in terms of roles of their alters.⁴ We have chosen potential descriptors for personal networks to cover as many indicators used by [Giannella and Fischer \(2016\)](#) as possible. Some variables they have applied are well represented in the SOEP data (e.g. the variables used to compute kin proximity), while others could not be meaningfully created due to a lack of data (e.g. homophily variables). In the following we present our selection of variables and

discuss how they relate to the dimensions of egocentric networks of [Giannella and Fischer \(2016\)](#).

While the Northern California Community Study (NCCS) made use of Fischer's (1982b) instrument for eliciting egocentric networks, the SOEP uses a variation of Burt's (1984) proposal, in which a total of five name generators with a maximum of five alters were used. In the SOEP, the smaller number of generators and the limitation of the naming of alteri lead to smaller networks with a high proportion of relatives, since family members are usually named first. In the SOEP data, essentially core networks were surveyed, whereas in the NCCS study, mainly wide networks with many weak ties were surveyed.

The five name generators of the SOEP are briefly explained because they form the basis for most of our variables. The name generators are questions about various support services provided by the alteri. In particular, egos were asked about (a) people, with whom they are discussing important things, (b) people who help with career advancement, (c) people who are asked for helping with long-term care, (d) with whom one occasionally has disputes, and (e) people who are allowed to tell the respondent unpleasant truths. Information on the relationship type or social context is available to describe the network persons. The relationship type is available for partners, family (e.g. mother/father, son/daughter, etc.) and relatives (mother-in-law/father-in-law, uncle/aunt, etc.). It represents the social role that this person has towards Ego. For family and kin relationships, there is also information on whether they are present at all (such as grandchildren or aunts/uncles), their number, and their residential distance from the respondent. With the help of the information on the presence of family members and their living distance, the living distance of the nuclear family was calculated, since needs for support services can also result from this. For friends and acquaintances, the network instrument provides information on the social context of the relationship, such as people from school/training/study or people from the neighborhood, etc. On the basis of this information, we formed variables that included the number of kin, nonkin and colleagues. We calculated IQV_role that shows how differently the networks are composed, in terms of kin, nonkin and colleague categories. We have also generated a variable that reflects the logarithm of the total number of relatives and a variable that indicates the total number of close friends. In addition, we computed the size of the networks accounting for redundancies, i.e. people having named their mother on all five name generators have only a network size of one. We also included the share of nonkin for every ego network. The SOEP data also provides information on whether the egos have lent money from kin or nonkin, which we also included.

Further variables were formed on the basis of the information on leisure behavior. This includes attending cultural events, such as concerts or theater, cinema, sports or dance events, one's own sporting activities, artistic and musical activities, socializing with friends, relatives or neighbors, helping friends or relatives, volunteering, participating in political parties or citizens' initiatives, and going to church

³ Since not every respondent was asked the questions of the network module and because we agreed on respondents with complete information, i.e., who had no missing data in their answers, the number of cases in our analyses amounts to 8,341. Although RF can handle missing data ([Shah et al., 2014](#)), we encountered problems in calculating capacity because RF uses $N \times N$ proximity matrices that grow in the order of N squares.

⁴ [Table 4](#) in the supplement shows descriptives of all chosen variables and provides detailed descriptions of operationalization.

or attending religious events. Although these variables do not directly measure contact to alters they at least indicate opportunities for meeting them (Giannella and Fischer, 2016: supplement page 1). Another variable reflects the number of organizational memberships of the respondents. Visits by and to persons of German origin as well as visits by and to persons of foreign origin also fall into this group of variables. An additional name generator also provides information on a maximum of three people who are important for the egos outside of their household. These were divided into kin and nonkin and the total number of persons mentioned was also recorded. Although some of the variables mentioned above include contact with both friends and relatives, we hoped to cover the dimensions of “nonkin interaction”, “activity-based”, and “religion-based”.

As already mentioned, it was also possible to form variables that represent the distance to relatives. The proportion of relatives who (a) live in the same place but not in their own household, (b) live further away and (c) live abroad were calculated. These variables serve to represent the dimension “kin proximity”. Finally, we have included the employment status and an indicator showing whether the respondents live in a partnership.

In this paragraph we would like to give a short overview of the analysis procedure proposed by Giannella and Fischer (2016). For those interested in a detailed description we strongly advise to take a look at their work. The suggested procedure to apply RF to derive an inductive typology of personal networks according to Giannella and Fischer (2016, 18 p.) consists of a total of 11 steps that are shortly described.

1. Choose variables that represent relevant aspects of egocentric networks.
2. Use RF in its unsupervised form to create an $N \times N$ proximity matrix. The values of this matrix represent closeness/similarity between the respondents.
3. Make use of the proximity matrix to cluster the data into a wide range of different partitions (10–50 clusters).
4. Apply RF to predict which respondent belongs to which clusters. Choose a number of clusters that have good prediction errors (<25)⁵ and treat them as being a good representation of the underlying structure of the data.
5. Reduce the observations to the number of respondents that belong to the well predicted clusters.
6. Cluster variables using the reduced data.
7. Combine variables to composite variables.
8. Use RF in unsupervised mode to compute a proximity matrix based on the composite variables only.
9. Use the proximity matrix to cluster the data again (2–20 clusters).
10. Apply RF to predict which observation belongs to which clusters and choose the best number of clusters.
11. Keep clusters with low prediction error ($<.4$) and more than 40 observations for the typology.

4. Dimensions of personal networks

As the data we used was not directly comparable to the NCCS data we tried numerous variations of data preparation e.g. including different variables or transforming them in different ways. Our analysis process revealed that steps 2 to 5 did not lead to any substantial differences in the subsequent clustering of the data as the clustering of our variables converged to the following results in almost every scenario. Therefore, we skipped these steps. Although Giannella and Fischer (2016) argued that these initial steps determine suitable variables as

separators, we could not find any significant changes whether we took those steps or not.

We then run the steps 6–11 proposed by Giannella and Fischer (2016) to build the typology. We first present the composite variables that were finally used to identify our inductive typology of egocentric networks. We also compare our findings to their results.

Twelve composite variables or dimensions of personal networks:

1. *Kin support* is a combination of the number of kin respondents named out of four name generators: discussing personal matters, help for career advancement, hypothetical help for long-term care of the respondent and talking about unpleasant truths. As the respondents could answer the name generators with a maximum of 5 alters, the composite variable ranges from 0–20. We decide to use this variant, which does not take into account whether the support comes from different people, since the network size already captures the number of different people in the network. Thus, the variables depict how many different support services respondents receive rather than the number of supporters.
2. *Nonkin support* is built analogous to kin support combining the number of nonkin named for these four generators. Nonkin, who are also employees or supervisors, were counted separately.
3. *Work support* is indexed by the number of coworkers and supervisors respondents named on the three name generators discussing personal matters, help for career advancement and talking about unpleasant truths.
4. Since we had contextual information about the named persons, we calculated a measure of the diversity of the network: Index of Qualitative Variation (IQV) provides information on the qualitative variation of relationship types in the network.⁶ The relationship types collected in the dataset (spouse (partner), former (spouse) partner, parent, child, grandparent, grandchild, relative, work colleague, educational person, neighborhood person, club person, recreational person, paid helper, and other person) represent different contexts in which respondents operate. The more diverse the types of relationships specified by the respondents, the more heterogeneous the composition of the network, or the greater the qualitative variation of social contexts in the network. Thus, the IQV index calculated here reflects the measure of homogeneity or heterogeneity of the respondents' social contexts in which they operate. We have referred to this measure as *IQV.role*, since the contexts represent the different roles in which the alteri with ego act.
5. *Social activities* measures the extent to which respondents reported helping friends and engaging in social activities with friends. Both questions were measured from 0–3 which results in a composite measure ranging from 0–6.
6. *Activities* combines whether respondents engage in sports, cultural events, art events, and visit movie theaters at least once a month.
7. *Organization-based* counts the number of organizational memberships and combines the count with engagement in honorary appointments and political engagement.
8. For the *network size*, we made sure that the number of mentions of family members did not exceed the number of mentions per category (sister, brother, etc.). If a person named a sister on four generators, but stated that he or she had only two sisters in total, only two people were used to calculate the network size. We did not have this information for friends and acquaintances. It is not possible to say, based on the data, whether or not five

⁵ The prediction error Giannella and Fischer (2016) used for their sample size.

⁶ The IQV was introduced into network analysis by Marsden (1987) for the analysis of egocentric network data following Agresti and Agresti (1978).

acquaintances named on five generators is always the same person. Therefore, we adjusted the number of persons to the number of acquaintances named per generator. This may theoretically underestimate the number of friends and acquaintances. Because of this limitation, we could not calculate multiplexity, neither for family members nor for friends and acquaintances.

9. *Proportion of nonkin* is the proportion of nonkin in the networks.
10. *Kin proximity* is the difference between near and far living kin. Close is defined as less than a 15-minute walk but not in the same household and far is everything over this distance except kin living abroad. Here, [Giannella and Fischer \(2016\)](#) form a measure from the number of relatives in the household, the percentage of kin living close by, and the percentage of kin living far away. Our variant thus focuses on the distance to kin who do not live in the same household.
11. *Kin abroad* is the proportion of kin living abroad.
12. *Church* is a single item that indicates whether respondents visit church at least once a month.

In comparison to [Giannella and Fischer \(2016\)](#) we can state that a high number of dimensions of personal networks could also be found in the SOEP. However, it must be pointed out that our composite variables only approximate those of [Giannella and Fischer \(2016\)](#), some better than others. The inclusion of individual variables, such as the IQV_role or the network size, is justified by their positive properties in the subsequent clustering of cases with the RF procedure. Although these variables can be assigned to one dimension according to the variable clustering, we found no convincing way to combine the different metrics.⁷ Simply adding up network size, IQV_role and the proportion of nonkin in the network would give different weightings to the individual variables which we wanted to avoid. The inclusion of the indicator for church attendance frequency is purely due to comparability with [Giannella and Fischer \(2016\)](#), since the variable clustering of this item would assign it to the “organization-based” dimension.

After clustering the variables, we used the RF procedure to create a similarity matrix which was then used for the final clustering of the ego networks. The results of these steps are presented in the following section.

5. Types of networks

As suggested by [Giannella and Fischer \(2016, 18–19 pp.\)](#) we interpreted the dimensions by normalizing their values to a range of 0 to 1. Accordingly, the cells of [Table 2](#) contain the average score of a cluster for the different dimensions. We interpreted the clusters by inspecting deviations of the cells from 1st and 3rd quartile values of the dimensions.

[Table 2](#) shows certain parallels between the typology of [Giannella and Fischer \(2016\)](#) and ours. If one groups the 14 types of our typology – as [Giannella and Fischer \(2016\)](#) did – into the categories “kin-dominant”, “nonkin-dominant”, and “small-network types”, similarities in the patterns of the individual dimensions can be discovered within these categories. Furthermore, we also found some distinctive new types. In the following, we will present the individual clusters, grouped according to the proposed categories and discuss parallels and differences to [Giannella and Fischer \(2016\)](#).

Kin-dominant types:

4. and 10. *Family and Community* (16%). Networks of this type are characterized in [Giannella and Fischer \(2016\)](#) by the fact that the dimension “kin involvement/support” has values beyond the third quartile. At the same time, however, there are also contacts to other areas of life, such as work, church or other activities. The Clusters 4 and

10 of our analyses show similar patterns. Kin support is strong and the dimensions organization-based, social activities and cultural activities are also moderate. Cluster 10 differs from Cluster 4 only in that it has smaller networks overall and the dimensions organization-based, social activities and cultural activities are somewhat weaker than in Cluster 4.

6. and 7. *Family only* (6%). This type is characterized by the fact that kin-involvement/ support has moderate values and at the same time very low values on the dimensions nonkin interaction and nonkin support are present ([Giannella and Fischer, 2016, 19 p.](#)). We see this type represented in Clusters 6 and 7 of our analyses as they are characterized by a focus on “kin”, although this focus is even clearer than in the clusters discussed earlier. The share of nonkin in the two clusters is 0 and no contacts to work colleagues are mentioned. In addition, the very low IQV_role points to a low level of mixing of these networks. This confirms the interpretation that the networks consist primarily of kin. Furthermore the values of the dimension “cultural activities” are at the lower end of the distribution and those of “church” are even 0.

5. and 14. *Home and Church* (12%). Clusters 5 and 10 have the characteristics of the “Home and Church” type ([Giannella and Fischer, 2016, 19 p.](#)). High values on the dimension “kin-support” with simultaneous absence of nonkin and very high values on the dimension “church”.

8. *Family centered and kin abroad* (3%). Cluster 8 is a type for which there is no equivalent in [Giannella and Fischer \(2016\)](#). It shows the highest proportion of kin living abroad while the proportion of close relatives is the second highest of all types. The focus of this type is clearly on kin and the connection to society via the dimensions social activities and cultural activities is relatively low.

Nonkin-dominant types:

2. *Career and friends* (28%). The career and friends cluster in [Giannella and Fischer \(2016\)](#) is characterized by a relatively low involvement with kin, relatively distant kin and shows a high proportion of nonkin and colleagues in the networks. Cluster 2 of our analyses shows clear parallels, except for the distance to the kin. Kin support is relatively low, while work support and nonkin support are above average. The proportion of nonkin in the networks also exceeds the 3rd quartile.

3. and 13. *Nonkin as kin* (6%). Clusters 3 and 13 show the highest values on the dimension nonkin-support. Likewise, kin support is comparatively low, although not quite as drastic as in the analyses by [Giannella and Fischer \(2016\)](#). The distance to kin is also average.

1. *Strongly embedded* (19%). For Cluster 1 of our analysis we could not find a suitable equivalent in [Giannella and Fischer \(2016\)](#). We refer to this cluster as “strongly embedded” because this type of network has relatively high values in almost all dimensions. This indicates a strong integration into society.

Small-network types:

11. *Withdrawn* (2%). As in [Giannella and Fischer \(2016\)](#) this type is characterized by many nearby kin. Although the kin support dimension in our analysis is at the maximum, the network size in combination with the IQV_role shows that the networks are very small and that there is only a small amount of people which get support.⁸

12. *Semi-isolated* (4%). As with [Giannella and Fischer \(2016\)](#), this type is characterized by the fact that relatively many kin live far away, but the available support comes exclusively from them. Dimensions such as the connection to organizations, social or cultural activities also show relatively low values.

9. *Small but heterogeneous* (6%). This type of network shows a similar pattern to the *Career and Friends* cluster but is well below the average

⁷ For a detailed description of the variable clustering see the supplement to this article.

⁸ The fact that the kin support has a maximum value here is due to the fact that we reported the dimensions as shares of the support across the three support dimensions in relation to the network size.

Table 2
Average scores on twelve dimensions for fourteen network types.

Cluster	N	kin support	nonkin support	work support	network size	proportion of nonkin	IQV_role
1 Strongly embedded	1550	.76	.16	.08	6.89	.52	.85
2 Career and friends	2312	.70	.22	.08	4.59	.62	.70
3 Nonkin focus	312	.73	.24	.03	4.00	.50	.81
4 Family and community	862	.88	.09	.03	4.84	.33	.79
5 Family and church	797	.92	.07	.01	3.64	<u>.22</u>	.65
6 Family only	271	1.00	<u>.00</u>	<u>.00</u>	<u>3.00</u>	<u>.00</u>	<u>.48</u>
7 Small family only	200	1.00	<u>.00</u>	<u>.00</u>	<u>1.59</u>	<u>.00</u>	<u>.26</u>
8 Family centered and kin abroad	243	1.00	<u>.00</u>	<u>.00</u>	<u>1.93</u>	<u>.00</u>	<u>.44</u>
9 Small but heterogeneous	509	.76	.20	.03	<u>2.01</u>	.50	<u>.54</u>
10 Small family and community	465	.85	.12	.03	3.01	.33	.72
11 Withdrawn	139	.99	<u>.00</u>	<u>.00</u>	<u>1.12</u>	<u>.00</u>	<u>.02</u>
12 Isolated	301	.99	<u>.00</u>	<u>.00</u>	<u>1.40</u>	<u>.00</u>	<u>.18</u>
13 Nonkin as kin	221	<u>.58</u>	.36	.06	3.01	.67	.72
14 Small family and church	159	1.00	<u>.00</u>	<u>.00</u>	<u>2.01</u>	<u>.00</u>	<u>.33</u>
1st quartile		.69	.00	.00	3.00	.25	.54
mean		.81	.14	.05	4.22	.41	.67
3rd quartile		1.00	.25	.07	5.00	.60	.81

Cluster	N	Δ kin near/far	proportion kin abroad	religious activity	orga-based	social activities	cultural activities
1 Strongly embedded	1550	–.32	<u>0.00</u>	.15	.15	.69	.35
2 Career and friends	2312	–.35	.01	.19	.12	.64	.30
3 Nonkin focus	312	–.37	.01	.14	.12	.59	.22
4 Family and community	862	–.32	.01	.09	.11	.65	.27
5 Family and church	797	–.27	.01	.25	.11	.64	.28
6 Family only	271	–.32	.01	<u>0.00</u>	.09	.58	.18
7 Small family only	200	–.29	.01	<u>0.00</u>	.09	1.00	.17
8 Family centered and kin abroad	243	.06	.13	.04	.06	<u>.38</u>	.12
9 Small but heterogeneous	509	–.31	.03	.13	.09	.57	.18
10 Small family and community	465	–.34	.01	.01	.09	.56	.21
11 Withdrawn	139	.37	.01	<u>0.00</u>	.06	<u>.24</u>	.08
12 Isolated	301	–.81	.01	<u>0.00</u>	.08	<u>.30</u>	.15
13 Nonkin as kin	221	–.44	.02	.01	.08	.60	.24
14 Small family and church	159	–.33	.04	1.00	.12	.63	.25
1st quartile		–.83	.00	.00	.00	.50	.00
mean		–.32	.02	.15	.11	.62	.26
3rd quartile		.00	.00	.00	.14	1.00	.50

Note: Averages for clusters on each of the 14 composite variables and single items.

Values above the 3rd quartile are in **bold** and values below the 1st quartile are underlined.

in terms of organization-based, social and cultural activities and the frequency of church visits, and the IQV_role is well below the value of the *Career and Friends* cluster. Persons of this network type, however, have smaller, partially heterogeneous networks, whereby the support by nonkin is apparently attributable to a few individuals.

Although we were able to find some similarities between our results and those of [Giannella and Fischer \(2016\)](#), it has certainly also become clear that there are definitely variations in the patterns found. We see two main possible explanations for this, the different data and the implementation of the method.

There are more than 30 years between the data we used and the NCCS data, during which various social processes, such as secularization or individualization, have progressed. In addition, the data come from different cultural backgrounds. Different patterns in the distributions of church attendance could be due to, for example, social change and/or different cultures (see clusters 6 and 7). Similarly, we have already drawn attention to the fact that the social network survey modules differ in the SOEP and NCCS. The different survey type results in a different composition of the networks, which is expressed, among other things, in a comparatively high proportion of relatives in the SOEP networks and a comparatively small network size. Against this background, one must ask to what extent the reported differences are due to this circumstance. Clusters 2, 3, and 13, in which nonkin-support is a crucial feature, could, for example, show clearer similarities if the SOEP networks contained more persons.

Last but not least, there remains the question of the influence of the implementation of the method on the results. Our analysis steps deviate at several points from those of [Giannella and Fischer \(2016\)](#), and the influence of these deviations remains unclear so far. In the conclusion,

we will revisit these and the previously mentioned differences and point out the need for further research.

6. Ego characteristics associated with network types

In order to demonstrate the sociological relevance of the uncovered clusters and to refine their interpretation, we show in this section descriptive characteristics of the individual clusters for selected attributes. In this way, we want to answer the question which characteristics the persons in the individual clusters have and to what extent these characteristics correspond to our interpretations. Since we have often combined two clusters into one type when comparing our clusters with those of [Giannella and Fischer \(2016\)](#), the representation of ego characteristics offers an opportunity to assess the practicability of this approach.

[Table 3](#) gives an overview of the distribution of ego characteristics. As with [Giannella and Fischer \(2016, 21 p.\)](#), the significance levels in [Table 3](#) indicate whether membership in a cluster is correlated with high or low values. For metric variables we used Pearson's *r* and for all other variables *Kendall's tau*. The following section provides a brief description of the variables used.

Kin-dominant types:

4. and 10. *Family and Community* (16%). [Table 3](#) shows clear differences when comparing the values of the combined Clusters 4 and 10. In Cluster 4 there is a correlation between cluster membership and gender, which is not the case in Cluster 10. Both clusters are made up of younger respondents, with persons assigned to Cluster 4 being on average even younger than those of Cluster 10. However, both clusters have similarly high values for partnership. There are

Table 3
Ego characteristics for fourteen network types.

	Cluster	Female	Age	Partner	Work_FT	Educ_years	Income	Mig_back
1	Strongly embedded	.56*	39.9***	.87***	.54***	13.3***	49,582***	.10***
2	Career and friends	.53	43.7***	.64***	.50+	12.6***	42,187+	.16+
3	Nonkin focus	.67***	49.1***	.91***	.40**	12.4	41,186	.13
4	Family and community	.63***	43.0***	.94***	.49	12.6*	45,630**	.13+
5	Family and church	.51+	42.9***	.82	.43**	12.4	44,080	.18*
6	Family only	.46*	49.4***	.96***	.46	11.9***	41,312	.15
7	Small family only	.38***	48.6***	.84***	.50	12.0*	36,040***	.15
8	Family centered and kin abroad	.51	49.9***	.88***	.36***	11.0***	36,456***	.36***
9	Small but heterogeneous	.49*	49.6***	.71***	.47	11.6***	39,199**	.21***
10	Small family and community	.55	48.4***	.93***	.50	11.9***	41,479	.11*
11	Withdrawn	.37***	51.5***	.79	.41+	10.8***	33,734***	.16
12	Isolated	.47*	54.8***	.89***	.44	11.8***	37,661***	.08**
13	Nonkin as kin	.51	46.3	.71***	.51	12.0*	38,098**	.11+
14	Small family and church	.53	58.4***	.87*	.30***	11.6***	39,488	.25***
	mean	.53	45.0	.80	.48	12.4	43,032	.15

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Significance levels indicate whether membership in a cluster is correlated with high or low values. For metric variables Pearson's r was used and for all other variables Kendall's τ .

also differences in income and education. Cluster 4 shows significantly higher values on both variables, while Cluster 10 has average income and even significantly lower education, which is certainly related to some extent. In comparison with the results of [Giannella and Fischer \(2016, 21 p.\)](#), the two clusters also show the highest values in terms of partnership and generate slightly higher or moderate incomes. In view of the differences listed, the question arises to what extent a merger of the two clusters can still be justified. The patterns of the personal networks of the two clusters may be comparable, but it turns out that the people are very different. This would suggest that the two clusters should be treated independently.

6. and 7. *Family only* (6%). For Clusters 6 and 7, which we have grouped under the Family only category, the differences in the ego characteristics are not so pronounced. Both clusters are characterized by a relatively high proportion of male respondents of average age. The only difference is the annual income, with Cluster 7 achieving significantly lower incomes on average. This type is not entirely comparable to Giannella and Fischer's "Family only" category. In their analysis the majority of egos were female (60%), only 64% had a stable partner, they were mostly not working fulltime and were economically disadvantaged ([Giannella and Fischer, 2016, 21 p.](#)). In our case more than 57% are male, 90% of them have a stable partner and their income is only slightly below average, with almost 50% working full time.

5. and 14. *Home and Church* (12%). While the cluster in [Giannella and Fischer \(2016, 21 p.\)](#) is mainly composed of women, this cannot be confirmed in our analysis. This difference can largely be attributed to the pronounced gender gap in religious attendance in the US, which is not even closely as pronounced in Germany ([PEW Research Center, 2016](#)). In our results, both clusters show no particular deviations from the average of the entire sample in terms of gender. However, there are again similarities with regard to low proportions of full-time employees. Cluster 14 even shows the lowest proportion of all clusters. It should also be noted here that there are significant differences between the two clusters. The years of education are positively correlated with Cluster 5, while there is a negative correlation with Cluster 14. This is also reflected in the incomes of the two clusters, although there are no significant correlations between cluster membership and income. Both are rather average or slightly below average. Overall, Cluster 14 seems to represent a small family and church-oriented network of mostly retired people of both genders. It shows the highest mean age and accordingly the lowest proportion of full-time workers of all clusters. It also shows the second highest proportion of egos with migration background. When the two clusters are blended the resulting type could be considered quite heterogeneous, with attendance of some kind of religious service as the primary uniting factor. Due to the big age difference of the two clusters it can be concluded that they represent different cohorts at different stages in the life cycle. This fact and the

difference in network size are arguments for a separate analysis of the two clusters.

8. *Family centered and kin abroad* (3%).⁹ This type is characterized by relatively low values for years of education and, associated with this, income. This cluster has the highest value for migration background. The respondents of this cluster are also relatively seldom full-time employees, which in turn offers another explanation for the low income. The interviewees are also somewhat older than average and to a large extent in a stable partnership. This type is a good example for the dependency of the typology on the respective variables used. [Giannella and Fischer \(2016\)](#) did not use any comparable variable to our kin abroad variable and therefore could not find a similar type.

Nonkin-dominant types:

2. *Career and friends* (28%). This type includes the lowest proportion of steady partnerships. Despite relatively high years of education, there is no clear significant correlation between this cluster and income; the average value of this cluster is even slightly below the average of the entire sample. On average it is one of the younger types albeit not as young as the one found in Giannella and Fischer's (2016) analysis. Otherwise, despite the use of different variables in our calculation, the resulting type seems similar to the one found by them.

3. and 13. *Nonkin as kin* (6%). This type contains two clusters that differ from each other in their ego characteristics. People in Cluster 3 are more likely to be women, to be slightly older than average, to be more likely to have a partner and to be less likely to work full-time. Education and income are largely in line with the average. For persons in Cluster 13, however, there is no correlation in terms of gender or age. However, they are less likely to have a partner and the education and income of the people in this cluster are also below average.

1. *Strongly embedded* (19%). The respondents in this cluster are on average the youngest, and the proportion of full-time employees is highest here. Education and income have the highest average values and there are positive correlations for living together with a spouse or partner.

Small-network types:

11. *Withdrawn* (2%). The respondents of this cluster are more likely to be male and their education is rather low. The average income of this cluster is also the lowest of all clusters. They are also slightly older than the average and less likely to have a steady partner. The nevertheless high proportion of those living with a partner (79%) and the network size of slightly above one point towards male retired egos, who live together with a female partner who is more socially embedded. This is in line with early findings of [Antonucci \(1985, 1994\)](#) and [Antonucci](#)

⁹ At first we suspected a network typical of migrants, because more than half of the persons reported that their country of origin is not Germany.

and Akiyama (1987a), who showed that older women have larger networks than men.

12. *Semi-isolated* (4%). People in this cluster are still slightly older than in the *Withdrawn* cluster, but also have lower education and income levels than the average. However, there is a positive correlation for living together with a spouse or partner and the proportion of men is not quite so high.

7. Discussion

Although the variables used in this analysis were measured differently, covering a much larger population at a different time and place, the results show comparability to Giannella and Fischer (2016). Moreover, the clusters exhibit sociologically meaningful differences. The basic dimensions of egocentric networks can be confirmed, which is surprising given the fact that the variables we used cover only core networks of respondents. The differentiators mentioned by Giannella and Fischer (2016) are also at play here and reveal basal similarities in the networks.

Nevertheless, the method used has problems from our point of view. Firstly, the results could only be achieved after many test runs. We walked through a “garden of forking paths” (Gelman and Loken, 2013). At each crossing decisions had to be made. Which respondents are included in the analysis? Which variables are chosen for the initial analysis of the basal dimensions of personal networks? How are these variables included in the statistical models? Which clustering algorithm will be used? The list of decisions to be taken is by no means complete. The small-step approach has many ways of influencing the results. For example, the question about the clustering algorithm was only answered regarding the results achieved and the interpretability of the clusters. By no means do we want to indicate that the results are erroneous, but we rejected results that were difficult to interpret or where the clusters could not be distinguished clearly enough.¹⁰ The impact of the many micro-decisions on the results should be further investigated. One possibility could be in so-called “multiverse analysis” (Steen et al., 2016) which aims at estimating the influence of the micro-decisions in order to get a better insight into their scope by basically computing every possible model.¹¹ In view of the time-consuming RF procedure and the vast number of possible models and our limited computing capacities, we have to leave this task to further research.

Secondly, the procedure can only provide substantial clustering regarding the used dimensions of social networks. This means that clusters that are defined, for example, by contacts to a religious community, are only revealed if a corresponding variable is used for clustering. This argument aims at the premise that this is an inductive typology. Hypothetically, further variables could be used for clustering, which could in turn yield different clustering results. Theoretical assumptions about what social networks could be about thus already form the results of the analysis when the variables are selected. For example, the cluster “family centered and kin abroad” can only be meaningfully revealed if the analysis contains a variable/dimension that covers this aspect of personal networks.

Even though the resulting typology is based on a vast amount of data, the data itself is in no way “neutral”. The amount of data may improve the validity of the typology, but there are countless micro-decisions which predetermine the outcome. Researchers have to be aware that every decision about the omission or inclusion of a variable affects the resulting typology.

¹⁰ The most common reason to reject a cluster solution was that the largest cluster was not subdivided by the algorithm and comprised almost half of the sample, which contradicted our goal of finding clusters that are small and easily distinguishable.

¹¹ We would like to thank Nico Sonntag and Dave Balzer for pointing out the possibility of a “multiverse analysis”.

8. Conclusion

The application of RF for a significantly larger dataset with limited network data can be considered successful in the sense that – after discarding some of the attempts – a differentiated, and meaningfully interpretable typology emerged. While resulting types share similar traits to the ones discovered by Giannella and Fischer (2016), one crucial aspect remains unclear: are the differences in the typology results of the application of the method, measurement disparities, e.g., diverging methods to elicit the respondent’s networks, or do they stem from distinct socio-demographic properties of the population? In some instances, like the gender gap in religious attendance in the US and the different age structure of the overall population which is subsequently mirrored in some of the clusters (e.g., the average age of a “young” cluster like “career and friends” is higher in Germany than in the US), the variation can be traced back to different socio-demographics. Not always is the root of the variation that clear. The next step in the creation of more stable types would be to systematically explain the differences between the two attempts at RF-based typologies, for example through a complete “multiverse analysis”. The consistency of results can only be thoroughly assessed when variables are omitted or included iteratively and systematically. Another fruitful approach could be to compare inductive clustering and traditional clustering methods within one dataset to directly test our assumption of more differentiated results of the former.

We believe that inductive clustering has the ability of uncovering correlations that have been going unnoticed before. We are under the impression, that such an explorative approach allows us to structure large datasets and fosters novel theoretical interpretations. Especially in network-oriented life course research an inductive empirics-theory link could be fruitful. Deductive approaches to clustering are often derived from very specific theories only concerned with a certain subpopulation (e.g., the elderly) leading to limited dimensionality of the network typology – usually drawing lines between family-, friend- and community orientation. It is exactly the openness of the inductive approach, the absence of preliminary theoretical considerations, that has the potential to empirically inform theory formation at this point in the research process. The number of clusters is not predetermined as when a deductive approach is used, and is depending entirely on their interpretability. Inductive typologies could be used as a heuristic to refine existing models, for example based on the life course cube (Bernardi et al., 2019) or on updated convoy models (Antonucci et al., 2019), further strengthening their network focus (Vacchiano and Spini, 2021; Settersten, 2015).

While our hopes of using RF as a procedure in compositional typology generation for large datasets were only partially fulfilled, there are some lessons to be learned from its application. The beforementioned drawbacks of the RF-approach could be turned into a potential advantage, when considered in the correct stage of the research process. Decisions in the “garden of forking paths” as well as about the inclusion or omission of variables in the application of the algorithm are inherently theory dependent. When these largely implicit theoretical assumptions can be made explicit, a debate about the validity of these decisions could follow. The countless micro-decisions underlying the application of data-mining algorithms must be addressed to ensure transparency and reproducibility of research. An open discourse about the theoretical underpinning of said micro-decisions could possibly result in a positive feedback loop, potentially linking inductive approaches, theory building or refinement and deductive attempts at falsification. In that sense the use of inductive methods can be seen as an interim step in a larger multidisciplinary attempt to improve the analysis of changing core networks across the life course.

To reach that goal, progress on methodological and theoretical grounds will not suffice. It should be accompanied by condensed empirical evidence. Prior research on network change over the life course has to be systematized and synthesized, preferably in longitudinal

studies covering the entire age range. While there is an abundance of sociological and psychological research on the change of egocentric networks during young adult life (Bidart, 2012; Degenne and Lebeaux, 2005) or on ecological transitions like entry into the labor market (Morrison, 2002), family formation (Bernardi, 2003; Bost et al., 2002; Kalmijn, 2003), partnership break-up (Zartler et al., 2015), death of the spouse (Zettel and Rook, 2004) and retirement (Cozijnsen et al., 2010), systematic overviews of network change across the life course are rare, the meta-analysis by Wrzus et al. (2013) and the longitudinal study by Mollenhorst et al. (2014) being notable exceptions. In the meta-analysis (Wrzus et al., 2013), a tendency of shrinking friend and stable family networks can be found across late adult life, confirming earlier findings (Antonucci and Akiyama, 1987b; Litwin, 1995b) and theoretical assumptions (Kahn and Antonucci, 1980). In our own results this general tendency of “network attrition” (Litwin, 1995b) is reflected in the two different “Family and church”-types, which apparently cover different phases in the life cycle — one consisting of mainly middle-aged parents with a bigger family-oriented network and higher income and education, and one consisting of retired people with a smaller family-only network and lower income. This example shows the potential of using RF in generating types that are more refined than before. With traditional clustering methods, usually four to six types of networks were found. Authors were distinguishing between family-centered, friend-centered, diverse and restricted networks (Fiori et al., 2006, 2008; Litwin, 2001; Park et al., 2018). Giannella and Fischer’s (2016) and our modest attempt show that a more detailed typology is possible, clearly linking certain types with different phases in the life cycle.

The possibility of compositional typology creation with more sophisticated methods also has implications for survey design. Availability of even more fine-grained network compositional data would in turn lead to a more detailed typology, potentially uncovering previously overlooked patterns. In an – admittedly optimistic – scenario this procedure could lead to an ongoing positive feedback loop of social network research oscillating between an explorative inductive, a theory-building and a hypothesis-testing deductive pole. Increasing data quality, methodological and theoretical advances would lead to a satisfactorily robust compositional typology of social networks, subsequently enabling a renewed framework which allows for a systematic network-based comparison of life course trajectories between different socio-demographic groups and populations.

9. Supplement

As suggested by Giannella and Fischer (2016), we began our analysis by determining appropriate variables to describe personal networks from the SOEP. Our aim was to find indicators for the known dimensions of the two authors, but also to cover possible further aspects of personal networks with the available data. In addition to the existing network module, which contains five name generators, other variables were used that only allow indirect conclusions about social contacts. We were limited to 2011 data because the network module is not included in every survey wave and only in that wave were certain variables included that seemed useful for describing personal networks. In the following we list all variables that were ultimately used in the RF procedure and describe their operationalization.

Explanation of selected variables

1. **partner** a dummy variable that indicates whether a respondent currently has a permanent partner or is married and lives with that person.¹²

- 2.–16. **kin, nonkin or work** preceding trust, career advancement, emergency care, quarrel, and unpleasant truths followed by the letter n indicates the number of kin, nonkin or coworkers that the respondent has indicated to the generator in question. Nonkin and coworker were distinguished from each other in order to have an approximation of contacts to the workplace, since the SOEP data otherwise did not contain any other suitable variables. If a person did not name any contacts, the category “nobody” was marked by a cross.
trust text: “With whom do you share personal thoughts and feelings or talk about things you wouldn’t tell everyone?”
career advancement text: “Who supports you in your professional advancement or education and helps you make progress?”
emergency care text: “Only hypothetically asked: What about a long-term need for care, e.g. after a serious accident: Who would you ask for help?”
quarrel text: “With whom do you occasionally have disputes or conflicts that are burdensome for you?”
unpleasant truth text: “Who can tell you unpleasant truths?”
- 17.–18. **visit_german** and **visit_foreign** are dummy variables that indicate whether respondents have had visits from Germans or foreigners in the last 12 months, or whether they have visited Germans or foreigners themselves.
- 19.–21. **kin_close**, **kin_far**, **kin_abroad** are the shares of relatives who live in the same place but under a 15 min walk, live further away or live abroad.
22. **log_kin_total** is the natural logarithm of the named kin on the five name generators. It was ignored that persons may have named the same kin on the name generators. This variable should represent the extent of available help, without considering from whom this support comes. We argue that it may be more helpful to have a lot of support, even if it is provided by only a few.
- 23.–25. **kin_3**, **nonkin_3** and **total_3** text: “Now a question about your circle of friends and acquaintances: Please think of three people outside your household who are important to you personally. They may be relatives or non relatives. Please tell us about the first, second and third person: Is he or she related to you?”. From the data on this question, the number of kin, nonkin and the total numbers were calculated.
26. **employ_status** represents the employment status of respondents, which was reported in 9 categories, e.g. full-time, part-time, educational or unemployed.
27. **orgs_involved** is an additive index of a total of five dummy variables. Memberships in trade unions, professional associations, personnel or works councils, environmental protection organizations and associations were surveyed.
- 28.–36. **cultural**, **movies/dance/concerts**, **sports**, **art**, **sociably**, **help**, **volunteer**, **politics**, and **church** are categorical variables with four values for the frequency of participation in the listed activities. The categories were recoded into dummy variables for which the value 1 indicates that a respondent performs the activity at least every month.
- 37.–40. **money_to_kin**, **money_to_nonkin**, **money_from_kin**, and **money_from_nonkin** variables that map financial support to kin and nonkin as well as from kin and nonkin.
41. **friends_total** is an open query on how many close friends the respondents have.
42. **IQV_role** The index of qualitative variation (IQV) is a measure of diversity, which provides information about the qualitative variation of the relationship types in the network. The IQV_role calculated here reflects the homogeneity or heterogeneity of the social contexts in the ego network.
43. **nw_size** is the doubling-cleaned size of personal networks. If, as in the example in the main text, only the mother is mentioned on each of the five name generators, the network size is 1.

¹² Same-sex partners are included.

Table 4
Network descriptors in the SOEP used in the Random Forest procedure.

	Descriptor	N	MEAN	SD	MIN	Pctl 25	MEDIAN	Pctl75	MAX
1	partner	8,341	0.80	0.40	0	1	1	1	1
2	kin_trust	8,341	2.08	1.28	0	1	2	3	5
3	nonkin_trust	8,341	0.47	0.71	0	0	0	1	5
4	work_trust	8,341	0.11	0.33	0	0	0	0	4
5	kin_career	8,341	0.85	1.11	0	0	0	1	5
6	nonkin_career	8,341	0.11	0.37	0	0	0	0	5
7	work_career	8,341	0.19	0.49	0	0	0	0	5
8	kin_emergency	8,341	1.92	1.27	0	1	2	3	5
9	nonkin_emergency	8,341	0.29	0.53	0	0	0	1	4
10	work_emergency	8,341	0.01	0.12	0	0	0	0	4
11	kin_quarrel	8,341	0.81	1.05	0	0	0	1	5
12	nonkin_quarrel	8,341	0.11	0.34	0	0	0	0	4
13	work_quarrel	8,341	0.13	0.39	0	0	0	0	4
14	kin_truth	8,341	1.81	1.32	0	1	2	3	5
15	nonkin_truth	8,341	0.32	0.57	0	0	0	1	5
16	work_truth	8,341	0.14	0.40	0	0	0	0	4
17	visit_german	8,341	0.97	0.17	0	1	1	1	1
18	visit_foreign	8,341	0.46	0.50	0	0	0	1	1
19	kin_close	8,341	0.22	0.29	0	0	0	0.4	1
20	kin_far	8,341	0.54	0.35	0	0.25	0.57	0.83	1
21	kin_abroad	8,341	0.05	0.15	0	0	0	0	1
22	log_kin_total	8,341	1.60	0.47	0	1.39	1.79	1.95	2.48
23	kin_3	8,341	1.18	1.02	0	0	1	2	3
24	nonkin_3	8,341	1.65	1.07	0	0	1	2	3
25	total_3	8,341	2.82	0.49	1	3	3	3	3
26	employ_status*	8,341							
27	orgs_involved	8,341	0.58	0.74	0	0	0	1	5
28	cultural	8,341	0.15	0.36	0	0	0	0	1
29	movies/dance/concert	8,341	0.25	0.43	0	0	0	0	1
30	sports	8,341	0.46	0.50	0	0	0	1	1
31	art	8,341	0.19	0.39	0	0	0	0	1
32	sociably	8,341	0.81	0.39	0	1	1	1	1
33	help	8,341	0.43	0.50	0	0	0	1	1
34	volunteer	8,341	0.19	0.40	0	0	0	0	1
35	politics	8,341	0.02	0.14	0	0	0	0	1
36	church	8,341	0.15	0.35	0	0	0	0	1
37	money_to_kin	8,341	35.31	861.80	0	0	0	0	51,000
38	money_to_nonkin	8,341	8.50	205.52	0	0	0	0	15,000
39	money_from_kin	8,341	8.15	217.06	0	0	0	0	15,000
40	money_from_nonkin	8,341	5.34	151.07	0	0	0	0	10,000
41	friends_total	8,341	4.37	3.74	0	2	4	5	60
42	IQV_role	8,341	0.67	0.22	0	0.54	0.72	0.81	0.95
43	nw_size	8,341	4.22	1.88	1	3	4	5	14
44	nw_nonkin	8,341	0.41	0.26	0	0.25	0.5	0.6	1

* employment status is measured in nine categories: 1. full-time (4,020 (48.20%));

2. part-time (1,222 (14.65%)); 3. vocational training (212 (2.54%)); 4. marginally employed (474 (5.68%));

5. Near retirement, zero working hours (49 (0.59%)); 6. Military service (3 (0.04%));

7. community service (19 (0.23%)); 8. sheltered workshop (16 (0.19%)); 9. not employed (2,326 (27.89%)).

44. **nw_nonkin** represents the proportion of nonkin in networks.

Analysis procedure

Originally we tried to reproduce the analyses of [Giannella and Fischer \(2016\)](#) as accurately as possible. However, since the process repeatedly revealed that steps 2–5 did not produce any noticeable differences in the grouping of the variables used, we did not take these steps and therefore began by grouping the variables in dimensions of egocentric networks.

Grouping Variables into dimensions

As with [Giannella and Fischer \(2016\)](#) we also used the R-package *clustOfVar* with the function *hclust()* to group the variables. In order to determine the number of variable clusters, we also used the *stability()* function of *clustOfVar* and used these results in combination with content criteria as a basis for decision-making. Considering these aspects, a solution of 12 clusters offers the best results from our point of view. [Table 5](#) provides a complete overview of the output of the *clustOfVar* method. In the formation of combining variables for the illustration of the different dimensions, we have essentially adhered to the specification of the *clustOfVar* procedure, whereby we have excluded items with factor loadings <0.30. In addition, we continued to use the **church** variable as a single variable. In [Giannella and Fischer](#)

(2016), religion-based represents a dimension of personal networks that we could not cover without the variable **church**. We have also kept the variables **IQV_role**, **nw_size** and **nw_nonkin** as single variables, since we do not know of any suitable way of combining the variables into an index without assigning them different weights. Since the Clusters 4 and 5 each contain only two items, each with factor loadings of the same size, we have used only one item for each of the further analysis steps to cover these dimensions. We also could not draw any clear conclusions from Clusters 6, 9, and 10, which is why we ignored them for the typologization.

Composite variables or dimensions of personal networks

1. *Kin support* is an additive index formed from the number of kin named on the name generators **trust**, **career**, **emergency**, and **truth**. The index has a range of 0–20, since a maximum of 5 people could be named for each name generator.
2. *Nonkin support* was formed analogously to the dimension kin support: **nonkin_trust** + **nonkin_career** + **nonkin_emergency** + **nonkin_truth**.
3. *Work support* was formed from the data for only three name generators: **work_trust**, **work_career**, and **work_truth** and therefore has a range of 0–15.

Table 5
Clustering of network descriptors with R-package *clustOfVar*.

Cluster 1		Cluster 6	
kin_emergency	.52	kin_3	.95
kin_trust	.50	nonkin_3	.95
kin_truth	.48	Cluster 7	
kin_career	.34	sociably	.51
log_kin_total	.27	help	.42
kin_quarrel	.15	friends_total	.28
Cluster 2		total_3	.21
nonkin_truth	.61	visit_german	.12
nonkin_trust	.59	Cluster 8	
nonkin_career	.34	volunteer	.60
nonkin_emergency	.31	orgs_involved	.46
nonkin_quarrel	.19	politics	.26
partner	.12	church	.25
Cluster 3		Cluster 9	
work_truth	.50	money_to_kin	.84
work_trust	.38	money_to_nonkin	.84
work_career	.37	Cluster 10	
employ_status	.30	money_from_kin	.72
work_quarrel	.24	money_from_nonkin	.72
work_emergency	.13	Cluster 11	
Cluster 4		IQV_role	.80
visit_foreign	.62	nw_size	.78
kin_abroad	.62	nw_nonkin	.53
Cluster 5		Cluster 12	
kin_close	.83	culture	.48
kin_far	.83	movies/dance/concert	.48
		sports	.38
		art	.33

4. *Index of qualitative Variation (IQV_role)* is a measure of diversity, which provides information about the qualitative variation of the relationship types in the network. In the case under examination here, it provides information about the distribution of relationships to the spouse (partner), former (spouse) partner, parents, children, grandparents, grandchildren, relatives, work colleagues, persons from education, people from the neighborhood, persons from a club, leisure, paid aids, outpatient service etc., and other persons. These types of relationships represent different contexts in which the respondents interact. The more varied the types of relations specified by the respondents, the more heterogeneous the composition of the network or, in other words, the greater the qualitative variation in the nominations in the network. The index of qualitative variation can vary from .00 – 1.00. “When all of the” named relationships of the “distributions are in one category, there is no diversity or variation, and the IQV is 0.00” (Crossman, 2021). Therefore, the IQV_role index calculated here reflects the homogeneity or heterogeneity of the social contexts in the ego network.
5. *Social activities* is a simple additive index of the two dummy variables **help** and **sociably**.
6. *Activities* is another additive index: **culture** + **movies/dance/concerts** + **sports** + **art**
7. *Organization-Based* An additive index of variables **orgs_involved**, **politics** and **volunteer**. Here we have refrained from transforming the variable **orgs_involved**, which has a range of 0–5, since it can be argued that the filling of an honorary office or participation in local politics have a similar significance as the organizational memberships from which **orgs_involved** was originally formed. The index thus ranges from 0–7 since **politics** and **volunteer** are coded binary.
8. *Network size* is the total number of different people named on 5 name generators. This means that if a person is named on several generators, he or she will still only be counted once.
9. *Proportion of nonkin* is the proportion of nonkin in egos networks, built from **nw_size**.
10. *Kin proximity* is the difference between near and far living kin, i.e. **kin_close** - **kin_far**. Close is defined as less than a 15-minute

walk but not in the same household and far is everything over this distance except kin living abroad.

11. *Kin abroad* is the proportion of kin living abroad. We choose to not combine this variable with **visit_foreign** as the two variables have different metrics and we did not want to just add them up.
12. *Church* is a single item that indicates whether respondents visit church at least once a month. We kept this variable as a single indicator for the dimension religion-based as the SOEP did not offer another suitable variable and we wanted to maintain a certain level of comparability to Giannella and Fischer (2016).

Unsupervised RF and clustering using the (composite) variables With the created composite variables, RF were used to create a dissimilarity matrix, which served as the basis for clustering. As with Giannella and Fischer (2016), our goal was to create a manageable number of clusters that were both sufficiently representative and interpretable. Following our primary source, we experimented with different settings of the RF procedure. For the presented solution we used Salford Predictive Modeler 8.2 with default settings and we only increased the number of trees to 1000. The composite variables were used as predictors with **church** being entered as categorical. For the subsequent clustering we used *agnes* (agglomerative nesting), which is implemented in the R-function *hclust*. With *agnes* we used *method=“ward”*, since this resulted in the smallest clusters in our analysis. Since RF is an inductive procedure, where many decisions are made during the procedure, we clearly recommend to use appropriate tests for the stability of the clustering procedure or the multiverse analysis mentioned in the article. Otherwise, it is quite possible that the results will not be reproducible. We recommend to use powerful computers for the execution of the procedure, because the computing time can take up to 5 h depending on the sample size.

Acknowledgments

Stefan Hundsdoerfer received a DOC-team fellowship funded by the Austrian Academy of Sciences (ÖAW) during the revision of the manuscript. The corresponding grant number is 70355 (DOC-team grant). We would like to express our gratitude to the anonymous reviewers. The appropriate but also benevolent criticism has always provided us with suggestions for new perspectives and thus initiated internal discussions that have helped the present article to achieve a higher quality.

References

- Agresti, A., Agresti, B.F., 1978. Statistical analysis of qualitative variation. In: Schuessler, K.F. (Ed.), *Sociological Methodology*. Jossey-Bass, San Francisco, pp. 204–237.
- Antonucci, T.C., 1985. Personal characteristics, social support, and social behavior. In: R.H., Binstock, Shanas, E. (Eds.), *Handbook of Aging and the Social Sciences*, second ed. Van Nostrand Reinhold, New York, pp. 94–128.
- Antonucci, T.C., 1994. A life-span view of women's social relations. In: Turner, B.F., Troll, L.E. (Eds.), *Women Growing Older*. Sage Publications, Thousand Oaks CA, pp. 239–269.
- Antonucci, T.C., Ajrouch, K.J., Webster, N.J., Zahodne, L.B., 2019. Social relations across the life span: Scientific advances, emerging issues, and future challenges. *Annu. Rev. Dev. Psychol.* 1, 313–336.
- Antonucci, T.C., Akiyama, H., 1987a. An examination of sex differences in social support among older men and women. *Sex Roles* 17 (11/12), 737–749.
- Antonucci, T.C., Akiyama, H., 1987b. Social networks in adult life and a preliminary examination of the convoy model. *J. Gerontol.* 42, 519–527. <http://dx.doi.org/10.1093/geronj/42.5.519>.
- Bernardi, L., 2003. Channels of social influence on reproduction. *Popul. Res. Policy Rev.* 22, 527–555. <http://dx.doi.org/10.1023/B:POPU.0000020892.15221.44>.
- Bernardi, L., Huinink, J., Settersten, R.A., 2019. The life course cube: A tool for studying lives. *Adv. Life Course Res.* 41, 1–13.
- Bidart, C., 2012. What does time imply? The contribution of longitudinal methods to the analysis of the life course. *Time Soc.* 22 (2), 254–273.
- Bidart, C., Degenne, A., Grossetti, M., 2018. Personal networks typologies: A structural approach. *Social Networks* 54, 1–11. <http://dx.doi.org/10.1016/j.socnet.2017.11.003>.

- Bost, K.K., Cox, M.J., Burchinal, M.R., Payne, C., 2002. Structural and supportive changes in couples' family and friendship networks across the transition to parenthood. *J. Marriage Fam.* 64, 517–531. <http://dx.doi.org/10.1111/j.1741-3737.2002.00517.x>.
- Breiman, L., 2001. Random forests. *Mach. Learn.* 45 (1), 5–32.
- Burt, R.S., 1984. Network items and the general social survey. *Social Networks* 6 (4), 293–339.
- Cozijnsen, R., Stevens, N.L., van Tilburg, T.G., 2010. Maintaining work-related personal ties following retirement. *Pers. Relatsh.* 17, 345–356. <http://dx.doi.org/10.1111/j.1475-6811.2010.01283.x>.
- Crossman, A., 2021. Index of qualitative variation. ThoughtCo., <https://www.thoughtco.com/index-of-qualitative-variation-iqv-3026700>. (Accessed 7 August 2021).
- Degenne, A., Lebeaux, M.O., 2005. The dynamics of personal networks at the time of entry into adult life. *Social Networks* 27, 337–358. <http://dx.doi.org/10.1016/j.socnet.2004.11.002>.
- Feld, S.L., 1981. The focused organization of social ties. *Am. J. Sociol.* 86 (5), 1015–1035.
- Fiori, K.L., Antonucci, T.C., Akiyama, H., 2008. Profiles of social relations among older adults: A cross-cultural approach. *Ageing Soc.* 28 (2), 203–231. <http://dx.doi.org/10.1017/S0144686X07006472>.
- Fiori, K.L., Antonucci, T.C., Cortina, K.S., 2006. Social network typologies and mental health among older adults. *J. Gerontol.: Ser. B* 61 (1), 25–32. <http://dx.doi.org/10.1093/geronb/61.1.P25>.
- Fischer, C.S., 1982a. To Dwell Among Friends. *Personal Networks in Town and City*. The University Press of Chicago, Chicago.
- Fischer, C.S., 1982b. What do we mean by friend? An inductive study. *Social Networks* 3 (4), 287–306.
- Freeman, L., Ruan, D., 1997. An international comparative study of interpersonal behavior and role relationships. *L. Année Sociologique* 47 (1), 89–115.
- Gelman, A., Loken, E., 2013. The garden of forking paths: Why multiple comparisons can be a problem, even when there is no fishing expedition or p-hacking and the research hypothesis was posited ahead of time. <http://www.stat.columbia.edu/gelman/research/unpublished/phacking.pdf>. (Accessed 8 July 2020).
- Giannella, E., Fischer, C.S., 2016. An inductive typology of egocentric networks. *Social Networks* 47, 15–23. <http://dx.doi.org/10.1016/j.socnet.2016.02.003>.
- Granovetter, M.S., 1973. The strength of weak ties. *Am. J. Sociol.* 78 (6), 1360–1380.
- Guadalupe, S., Vicente, H.T., 2020. Types of personal social networks of older adults in Portugal. <http://dx.doi.org/10.1007/s11205-019-02252-3>, *Social Indicators Research* (Web).
- Hlebec, V., Kogovšek, T., 2013. Different approaches to measure ego-centered social support networks: A meta-analysis. *Qual. Quant.* 47, 3435–3455. <http://dx.doi.org/10.1007/s11135-012-9731-2>.
- Kahn, R.L., Antonucci, T.C., 1980. Convoys over the life course: Attachment, roles and social support. In: Baltes, P.B., Brim, O. (Eds.), *Life-Span Development and Behavior*. Academic Press, New York, pp. 253–286.
- Kalmijn, M., 2003. Shared friendship networks and the life course: An analysis of survey data on married and cohabiting couples. *Social Networks* 25 (3), 231–249.
- Kogovšek, T., Coenders, G., Hlebec, V., 2013. Predictors and outcomes of social network compositions: A compositional structural equation modeling approach. *Social Networks* 35, 1–10.
- Lin, N., 1999. Building a network theory of social capital. *Connections* 22 (1), 8–51.
- Litwin, H., 1995a. The social networks of elderly immigrants: An analytic typology. *J. Aging Stud.* 9 (2), 155–174. [http://dx.doi.org/10.1016/0890-4065\(95\)90009-8](http://dx.doi.org/10.1016/0890-4065(95)90009-8).
- Litwin, H., 1995b. Uprooted in Old Age: Soviet Jews and their Social Networks in Israel. Praeger, Westport, CT.
- Litwin, H., 2001. Social network type and morale in old age. *Gerontologist* 41, 516–524. <http://dx.doi.org/10.1093/geront/41.4.516>.
- Marsden, P.V., 1987. Core discussion networks of Americans. *Am. Sociol. Rev.* 52 (1), 122–131.
- Maya-Jariego, I., 2021. Building a structural typology of personal networks: Individual differences in the cohesion of interpersonal environment. *Social Networks* 64, 173–180.
- McCarty, C., Lubbers, M.J., Vacca, R., Molina, J.L. (Eds.), 2019. *Conducting Personal Network Research: A Practical Guide*. Guilford Publishers, New York.
- Mollenhorst, G., Volker, B., Flap, H., 2014. Changes in personal relationships: How social contexts affect the emergence and discontinuation of relationships. *Social Networks* 37, 65–80.
- Morrison, E.W., 2002. Newcomers' relationships: The role of social network ties during socialization. *Acad. Manag. J.* 45, 1149–1160. <http://dx.doi.org/10.2307/3069430>.
- Park, N.S., Chiriboga, D.A., Chung, S., 2018. A typology of social networks and its relationship to psychological well-being in Korean adults. *Int. J. Aging Hum. Dev.* 90 (3), 211–233. <http://dx.doi.org/10.1177/0091415018815230>.
- Park, N.S., Jang, Y., Lee, B.S., Ko, J.E., Haley, W.E., Chiriboga, D.A., 2015. An empirical typology of social networks and its association with physical and mental health: A study with older Korean immigrants. *J. Gerontol. Ser. B* 70 (1), 67–76. <http://dx.doi.org/10.1093/geronb/gbt065>.
- PEW Research Center, 2016. The gender gap in religion around the world. 6. In the US, Religious Commitment is High and the Gender Gap is Wide. Pew Research Center, <https://www.pewforum.org/2016/03/22/the-gender-gap-in-religion-around-the-world/>. (Accessed 8 July 2020).
- Settersten, R.A., 2015. Relationships in time and the life course: The significance of linked lives. *Res. Hum. Dev.* 12, 217–223.
- Shah, A.D., Bartlett, J.W., Carpenter, J., Nicholas, O., Hemingway, H., 2014. Comparison of random forest and parametric imputation models for imputing missing data using MICE: A CALIBER study. *Am. J. Epidemiol.* 179 (6), 764–774. <http://dx.doi.org/10.1093/aje/kwt312>.
- Socio-Economic Panel (SOEP), 2020. Data for years 1984–2016. <http://dx.doi.org/10.5684/soep.v33>, Version 33, SOEP.
- Steege, S., Tuerlinckx, F., Gelman, A., Vanpaemel, W., 2016. Increasing transparency through a multiverse analysis. *Perspect. Psychol. Sci.* 11 (5), 702–712.
- Szabo, A., Stephens, C., Allen, J., Alpass, F., 2018. Construct validation of Wenger's support network typology. *Gerontol. B Psychol. Sci. Soc. Sci.* 73 (7), 1244–1249. <http://dx.doi.org/10.1093/geronb/gbw126>.
- Vacca, R., 2020. Structure in personal networks: Constructing and comparing typologies. *Netw. Sci.* 8 (2), 142–167. <http://dx.doi.org/10.1017/nws.2019.29>.
- Vacchiano, M., Spini, D., 2021. Networked lives. *J. Theor. Soc. Behav.* 51 (1), 87–103.
- Wagner, G.G., Frick, J.R., Schupp, J., 2007. The German Socio-Economic Panel Study (SOEP) – Scope, evolution and enhancements. *SOEP Papers on Multidisciplinary Panel Data Research*, (1), Deutsches Institut für Wirtschaftsforschung (DIW), Berlin.
- Wenger, G.C., 1989. Support networks in old age — Constructing a typology. In: Jeffreys, M. (Ed.), *As Britain Ages*. Routledge, London, pp. 166–185.
- Wenger, G.C., 1991. A network typology: From theory to practice. *J. Aging Stud.* 5, 147–162. [http://dx.doi.org/10.1016/0890-4065\(91\)90003-B](http://dx.doi.org/10.1016/0890-4065(91)90003-B).
- Wenger, G.C., Shahtahmasebi, S., 1991. Survivors: Support network variation and sources of help in rural communities. *J. Cross Cult. Gerontol.* 6 (1), 41–82. <http://dx.doi.org/10.1007/bf00117111>.
- Wolf, C., 2010. Egozentrierte netzwerke: Datenerhebung und datenanalyse. In: Stegbauer, C., Häußling, R. (Eds.), *Handbuch Netzwerkforschung*. VS Verlag, Wiesbaden, pp. 471–483.
- Wrzus, C., Hänel, M., Wagner, J., Neyer, F.J., 2013. Social network changes and life events across the life span: A meta-analysis. *Psychol. Bull.* 139 (1), 53–80. <http://dx.doi.org/10.1037/a0028601>.
- Zartler, U., Heintz-Martin, V., Arránz Becker, O. (Eds.), 2015. *Family Dynamics After Separation: A Life Course Perspective on Post-Divorce Families*. Verlag Barbara Budrich, Leverkusen-Opladen.
- Zettel, L.A., Rook, K.S., 2004. Substitution and compensation in the social networks of older widowed women. *Psychol. Aging* 19, 433–443. <http://dx.doi.org/10.1037/0882-7974.19.3.433>.